④



$$G(s) = \frac{s+12}{s+75}$$

$$H(s) = \frac{s+10}{s+2}$$

a) $= \frac{G(s)}{1+H(s)G(s)} = \frac{\left(\frac{s+12}{s+75}\right)}{1+\left(\frac{s+12}{s+75}\right)\left(\frac{s+10}{s+2}\right)} \cdot \frac{(s+75)(s+2)}{(s+75)(s+2)}$

$= \frac{(s+12)(s+2)}{(s+75)(s+2)+(s+12)(s+10)} = \frac{s^2+14s+24}{(s^2+77s+150)+(s^2+22s+120)}$

$= \frac{s^2+14s+24}{2s^2+99s+270}$

b) Both the numerator and denominator are to the 2nd degree ∴ This is not strictly proper (this took a bit)

$14-\frac{99}{2}=-35.5 \quad 24-\frac{270}{2}=-111$

$\frac{s^2+14s+24}{2s^2+99s+270} = \left[\frac{-35.5s-111}{2s^2+99s+270}\right] + 1/2$

d) $\frac{-35.5s - 111}{s^2 + 99s + 270} = \frac{-12.75s - 55.5}{s^2 + 49.5s + 135}$

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -135 & -49.55 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -135 & -49.55 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = [-55.5 \quad -17.75] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [1/2] u$$

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c) $\frac{-17.75s - 55.5}{s^2 + 49.55 + 135}$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & -135 \\ 1 & -49.55 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} -55.5 \\ -17.75 \end{bmatrix} u$$

$$y = [0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [1/2] u$$

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e) simulink